JAYCE HILL

Principal AI Security Architect | Agentic AI & MCP Security · Threat Modeling · Cloud Security Architecture

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EXECUTIVE SUMMARY

Principal security architect with 20+ years designing and delivering security solutions for Fortune 100 enterprises, the Department of Defense, and the Intelligence Community. At Microsoft ISE, operated as a customer-facing forward deployed engineer: co-authored and co-led **CRISP**, an MCP-based threat modeling framework deployed across 100+ enterprise customers globally, and personally led 20 engagements from discovery through delivery. Security lead on **HVE-Core**, a 34-agent AI engineering platform. Builds reference architectures, labs, and demos, including an open-source **secure MCP execution runtime**, and is equally comfortable whiteboarding with a CISO or working hands-on with an AI/ML engineering team.

CORE COMPETENCIES

- AI Infrastructure Security (AKS, AI Foundry, pipelines)
- AI Application Security (LLM, RAG, agentic workflows)
- Public-Facing AI & API Security
- Prompt Injection, Data Poisoning & Excessive Agency
- Reference Architectures & Demo Environments
- Technical Discovery & Proof-of-Concept Delivery
- CISO & Executive Presentations
- Cloud Security Posture (Defender for Cloud, policy-as-code)
- Threat Modeling (CRISP, STRIDE, MITRE ATT&CK)
- Software Supply Chain (SBOM, SLSA, Sigstore)
- Zero Trust & Identity Governance
- Field Enablement, Workshops & Hackathons

TECHNICAL SKILLS

AI Security: OWASP Top 10 for LLM, MCP security proxy, OPA/Rego policy, Entra Agent ID, gVisor sandbox design, Defender for AI

Cloud & Posture: Microsoft Azure (Solutions Architect Expert), AI Foundry, AKS, Defender for Cloud, Sentinel; situational AWS and OCI

Frameworks: NIST AI RMF, Google SAIF, MITRE ATT&CK, NIST 800-53, SLSA, OpenSSF Scorecard, Zero Trust

Languages / IaC: Python, PowerShell, Bash, Terraform, Bicep

PROFESSIONAL EXPERIENCE

Principal Security Software Engineer | Microsoft, Industry Solutions Engineering *2021 – 2026*

Forward Deployed Engineer embedded with Fortune 100 and public sector customers; Security Lead for ISE's agentic AI engineering platform. Individual contributor, Level 65 (Principal).

- Co-authored and co-led **CRISP**, a threat modeling framework operating as an MCP server that scans customer architectures, generates data attribute tables, and produces security plans; deployed across 100+ enterprise customers in EMEA, APAC, and the Americas and named a top-priority accelerator by ISE leadership.
- Personally led on-the-ground CRISP deployment for 20 engagements, including Dow, Halliburton, LSEG, Bank of America, Fidelity, General Motors, KPMG, and FIS; at Halliburton, combined CRISP with HVE-Core to regenerate and validate security plans as architecture evolved weekly, cutting the planning cycle from weeks to about an hour per iteration while preserving human-in-the-loop oversight.
- Security lead on **HVE-Core**, a 34-agent AI engineering platform: co-designed the access-control architecture, owned the security review, and red-teamed agents to enforce human-in-the-loop and accuracy controls; ran deployment workshops at KPMG and LSEG and presented at LSEG's HVE Hackathon Science Fair.
- Conducted architecture and code reviews across engagements, driving recurring findings to root cause and feeding fixes back into CRISP's standardized frameworks; translated Defender for Cloud posture findings into policy-as-code deployed in customer environments.
- Authored a secure-by-default Bicep library spanning 11 Azure services (AKS, SQL, Cosmos DB, Key Vault, Redis, AI Foundry, Storage, VNet, NSG, PIM) with Zero Trust defaults, and hardened CRISP's CI/CD and software supply chain (CodeQL SAST, GitHub Actions SHA pinning, Trivy, CycloneDX SBOM signing) to OpenSSF Scorecard standards.

Information System Security Manager | General Dynamics IT (GDIT) *2020 – 2021*

Security accreditation and operations for classified cloud systems supporting IC missions.

- Owned security accreditation (ATO) and Continuous Monitoring under RMF for classified cloud-based systems on AWS, engaging directly with the Authorizing Official and AO-designated representatives to move authorization packages through the RMF lifecycle.

- Conducted architecture reviews and Technical Exchange Meetings with program leadership to assess risk posture, define controls, and align remediation milestones.
- Led post-incident analysis for data spills and drove zero-day vulnerability resolution.

Senior Principal Consultant, National Security Group | Oracle *2011 – 2020*

Built and implemented offensive and defensive cyber capabilities for DoD customers over a 9-year engagement.

- Led MITRE ATT&CK-based penetration testing, purple team operations, and vulnerability assessment across Oracle Exadata, Big Data Appliance, WebLogic, and Analytics platforms, validating adversary tradecraft against DoD mission systems.
- Served as Oracle Cloud Infrastructure Solution Architect, designing deployment architectures and leading certification and accreditation for national security programs; translated technical findings into program- and executive-level risk posture.
- Facilitated quarterly stakeholder reviews and bi-monthly Technical Exchange Meetings on risk, emerging technology, and program milestones.

Systems Security Engineer | Booz Allen Hamilton *2008 – 2011*

- Performed certification and accreditation, risk assessments, and security control validation for NSA mission systems under DCID 6/3, coordinating findings with government stakeholders and program security officers.

Earlier Career | NJVC · L-3 · Lockheed Martin · Titan Systems *2003 – 2008*

- Systems and security engineering roles supporting national security and defense programs.

INDEPENDENT RESEARCH & HANDS-ON LABS

Secure MCP Execution Runtime — Open-Source Reference Architecture | Hill Security Advisors *2026*

- Built and open-sourced a secure execution runtime for agentic AI tool calls: a Python JSON-RPC security proxy with recursive argument sanitization and structured telemetry, a proof-of-possession (RFC 7800) token validator for short-lived agent identities, and a default-deny OPA/Rego tool-authorization policy, with unit tests and CodeQL running in CI.
- Delivered Azure IaC for the control plane (Entra Agent ID, APIM inspection policy, KQL audit queries), a threat matrix mapping prompt injection, excessive agency, and non-human identity risk to OWASP Top 10 for LLM, Google SAIF, and NIST AI RMF, and a gVisor execution-sandbox design; code at github.com/jaycehilleng/jaycehilleng.github.io.

OpenClaw Agentic AI Lab | Hill Security Advisors *2026*

- Building and security-reviewing OpenClaw, an open-source agentic AI assistant, on an isolated Azure VM in a dedicated lab tenant; designing exec-approval routing and human-in-the-loop gating before enabling autonomous tool execution.

EDUCATION

M.S. Cybersecurity (in progress) — Georgia Institute of Technology
M.Eng., Systems Engineering — George Washington University
B.S., Electrical & Computer Engineering — Michigan State University

CERTIFICATIONS & CLEARANCE

Microsoft Azure Solutions Architect Expert · GIAC Security Essentials (GSEC) · Certified Ethical Hacker (CEH)
TS/SCI — Active (Full Scope Polygraph held continuously 2003–2023)

COMMUNITY LEADERSHIP

100 Black Men of Greater Washington, DC: Vice President of Operations and Development (2015–2019); Board of Directors (2011–2019). Created STEM The 100 Way, impacting 3,000+ students, including an ongoing STEM partnership with the Department of Defense.